GATE ME 90 Days Challenge

Day 30/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Weekly Revision	Week 5 Revision	77 days to GATE

How to use this set: 15 questions (5 MCQ, 10 NAT) on all five topics of Week 5: riveted, bolted and welded joints; machine elements; vector calculus; fluid properties and statics; fluid kinematics and Bernoulli (3 questions each). Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.

Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 NAT

A double-strap butt joint has one rivet per pitch length in double shear. Plate thickness = 12 mm, rivet-hole diameter = 20 mm, pitch = 70 mm. Permissible stresses: tension 90 MPa, shear 75 MPa, crushing 150 MPa. Taking the double-shear area as twice the single-shear area, the efficiency of the joint (in %, round off to two decimal places) is _____

Q2 MCQ

A plate is joined to another by two parallel fillet welds of 10 mm leg size and carries a load of 80 kN along the welds. If the permissible shear stress in the weld is 56 MPa, the required length of each weld (neglecting end allowances) is about

(A) 71.4 mm
(B) 202.1 mm
(C) 101.0 mm
(D) 142.9 mm

Q3 NAT

In a single-riveted lap joint, the rivet-hole diameter is 20 mm and the plates are 10 mm thick. Permissible stresses: tension 80 MPa, shear 60 MPa. The pitch (in mm, round off to two decimal places) at which the tearing strength of the plate equals the shearing strength of the rivet is _____

Q4 NAT

A cone clutch has a semi-cone angle of 12.5°, a mean diameter of 250 mm and $\mu = 0.3$. The axial force is 1 kN. Using the uniform-wear theory, the torque it can transmit (in N·m, round off to one decimal place) is _____

Q5 MCQ

The minimum number of teeth on a 20° full-depth involute pinion (addendum = 1 module) that meshes with a rack without interference is

(A) 17
(B) 18
(C) 32
(D) 14

Q6 NAT

A helical compression spring is made of 8 mm wire with a mean coil diameter of 48 mm and carries an axial load of 500 N. Using Wahl's factor, the maximum shear stress in the wire (in MPa, round off to one decimal place) is _____

Q7 NAT

The vector field $\mathbf{F} = (x + 3y)\,\mathbf{i} + (y - 2z)\,\mathbf{j} + (x + pz)\,\mathbf{k}$ is solenoidal. The value of p is _____

Q8 MCQ

Using Stokes' theorem, the value of $\oint_C (z\,dx + x\,dy + y\,dz)$, where C is the circle $x^2 + y^2 = 1$, $z = 0$, traversed anticlockwise when seen from above, is

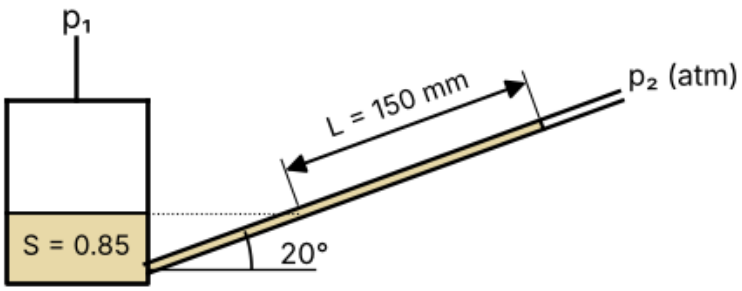
(A) 2π **(B)** π
(C) 0 **(D)** 3π

Q9 NAT

The acute angle between the surfaces $x^2 + y^2 + z^2 = 9$ and $z = x^2 + y^2 - 3$ at the point $(2, -1, 2)$ (in degrees, round off to two decimal places) is _____

Q10 NAT

In the inclined-tube manometer shown, the tube is at 20° to the horizontal and the reservoir area is 100 times the tube area. The gauge liquid has S = 0.85. When pressure p_1 is applied to the reservoir, the meniscus moves L = 150 mm along the tube from its zero mark. The pressure difference $p_1 - p_2$ (in Pa, round off to one decimal place) is _____



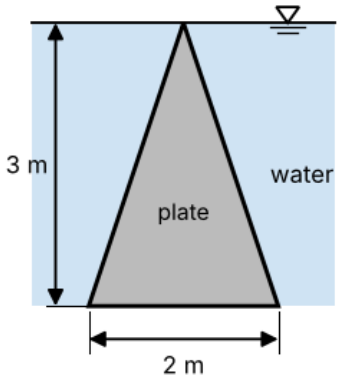
Q11 MCQ

An open rectangular tank partly filled with water moves horizontally with a constant acceleration of 4.905 m/s². The angle the free surface makes with the horizontal is

(A) 30.00° **(B)** 45.00°
(C) 63.43° **(D)** 26.57°

Q12 NAT

A triangular plate with base 2 m and height 3 m is immersed vertically in water with its apex at the free surface and its base horizontal, as shown. The depth of the centre of pressure below the free surface (in m, round off to two decimal places) is _____



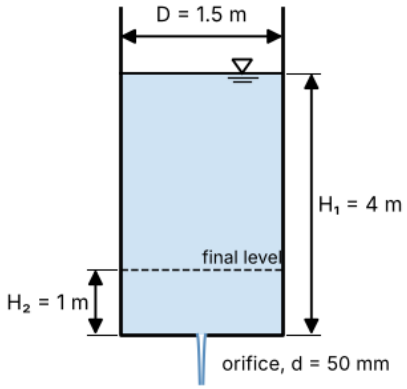
Q13 MCQ

The velocity potential of a 2-D flow is $\varphi = x^2 - y^2$. With $u = \partial\varphi/\partial x = \partial\psi/\partial y$ and $v = \partial\varphi/\partial y = -\partial\psi/\partial x$, the stream function ψ (taking $\psi = 0$ at the origin) is

(A) $2xy$
(B) $-2xy$
(C) $x^2 + y^2$
(D) xy

Q14 NAT

An open vertical cylindrical tank of 1.5 m diameter drains through a 50 mm diameter orifice in its base ($C_d = 0.62$), as shown. The time for the water level to fall from 4 m to 1 m above the orifice (in s, round off to the nearest whole number) is _____



Q15 NAT

A steady 2-D velocity field is $\mathbf{V} = 2xy\,\mathbf{i} - y^2\,\mathbf{j}$ (m/s, x and y in m). The magnitude of the fluid acceleration at the point (1, 2) (in m/s², round off to two decimal places) is _____

Answer Key

Q	Type	Answer
Q1	NAT	47.62 (accept 47.33 to 47.90)
Q2	MCQ	(C) 101.0 mm
Q3	NAT	43.56 (accept 43.30 to 43.82)
Q4	NAT	173.3 (accept 172.2 to 174.3)
Q5	MCQ	(B) 18
Q6	NAT	149.5 (accept 148.6 to 150.4)
Q7	NAT	−2
Q8	MCQ	(B) π
Q9	NAT	54.41 (accept 54.31 to 54.52)
Q10	NAT	440.3 (accept 437.7 to 442.9)
Q11	MCQ	(D) 26.57°
Q12	NAT	2.25 (accept 2.24 to 2.26)
Q13	MCQ	(A) 2xy
Q14	NAT	655 (accept 648 to 662)
Q15	NAT	17.89 (accept 17.82 to 17.96)

Detailed Solutions

Q1 **Answer: 47.62 (accept 47.33 to 47.90)**

Concept: Find the tearing, shearing and crushing strengths per pitch; the least of them divided by the strength of the solid plate is the efficiency.

Tearing: $(p - d)t\sigma_t = 50 \times 12 \times 90 =$ **54.0 kN**

Shearing (double): $2 \times (\pi/4)(20^2) \times 75 =$ **47.12 kN**

Crushing: $d\ t\ \sigma_c = 20 \times 12 \times 150 =$ **36.0 kN** (least)

Solid plate: $p\ t\ \sigma_t = 70 \times 12 \times 90 = 75.6$ kN, so $\eta = 36.0/75.6 =$ **47.62 %**

Exam tip: Many codes take the double-shear area as 1.875 × single shear; use whatever factor the question states.

Q2 **Answer: (C) 101.0 mm**

Concept: A parallel fillet weld fails in shear across its throat, t = 0.707 s.

$P = 2 \times 0.707\ s\ l\ \tau \rightarrow l = 80\,000/(2 \times 0.707 \times 10 \times 56) =$ **101.0 mm**

Add about 12.5 mm to each weld for starting and stopping in practice.

Exam tip: Forgetting the 0.707 gives 71.4 mm, an unsafe answer; a single weld would need 202 mm.

Q3 **Answer: 43.56 (accept 43.30 to 43.82)**

Concept: Equate the two strengths per pitch length: $(p - d)\, t\, \sigma_t = (\pi/4)\, d^2\, \tau$.

$(\pi/4)(20^2)(60) =$ **18850 N**

$p - 20 = 18\,850/(10 \times 80) = 23.56$ mm → $p =$ **43.56 mm**

Exam tip: A larger pitch makes the rivet weaker than the plate; a smaller one makes the plate weaker. Equal strengths give the best design.

Q4 **Answer: 173.3 (accept 172.2 to 174.3)**

Concept: For a cone clutch, $T = \mu\, W\, R_m/\sin\alpha$ (uniform wear, R_m = mean radius).

$T = 0.3 \times 1000 \times 0.125/\sin 12.5° = 37.5/0.2164 =$ **173.3 N·m**

Exam tip: The $1/\sin\alpha$ factor (= 4.62 here) is the wedge action; it is why small cone angles give large torque but can make the clutch hard to disengage.

Q5 **Answer: (B) 18**

Concept: For a rack, interference is avoided when the pinion has at least $z = 2a_w/\sin^2\varphi$ teeth, a_w = addendum in modules.

$z = 2 \times 1/\sin^2 20° = 2/0.1170 = 17.10$

The number of teeth must be a whole number not below this, so **18**.

Exam tip: Rounding down to 17 is the trap. For 14.5° full depth the answer is 32.

Q6 **Answer: 149.5 (accept 148.6 to 150.4)**

Concept: $\tau = K \times 8PD/(\pi d^3)$, with Wahl's factor $K = (4C - 1)/(4C - 4) + 0.615/C$ accounting for direct shear and curvature.

$C = D/d = 6$, $K = 23/20 + 0.615/6 =$ **1.2525**

$\tau = 1.2525 \times 8 \times 500 \times 48/(\pi \times 8^3) =$ **149.5 MPa**

Exam tip: Without Wahl's factor the stress would be only 119.4 MPa; springs of small index C need the full correction.

Q7 **Answer: −2**

Concept: A field is solenoidal when its divergence is zero everywhere.

$\nabla\cdot\mathbf{F} = 1 + 1 + p = 0 \rightarrow$ **p = −2**

Exam tip: Solenoidal = divergence-free (no sources or sinks); irrotational = curl-free. Do not mix them up.

Q8 **Answer: (B) π**

Concept: $\oint \mathbf{F}\cdot d\mathbf{r} = \iint (\nabla\times\mathbf{F})\cdot\mathbf{n}\, dS$ over any surface bounded by C.

$\mathbf{F} = (z, x, y) \rightarrow \nabla\times\mathbf{F} = (1, 1, 1)$. Take the flat unit disc, $\mathbf{n} = \mathbf{k}$.

$\iint (\nabla\times\mathbf{F})\cdot\mathbf{k}\, dS = \iint 1\, dS =$ area = **π**

Exam tip: Check directly: $x = \cos t$, $y = \sin t$, $z = 0$ gives $\int \cos^2 t\, dt$ over 0 to $2\pi = \pi$.

Q9 **Answer: 54.41 (accept 54.31 to 54.52)**

Concept: The angle between two surfaces at a point is the angle between their normals, i.e. their gradients.

$\nabla(x^2 + y^2 + z^2) = (2x, 2y, 2z) = (4, -2, 4)$; $\nabla(x^2 + y^2 - z) = (2x, 2y, -1) = (4, -2, -1)$

$\cos\theta = (16 + 4 - 4)/(6 \times \sqrt{21}) = 16/27.50 = 0.5819 \rightarrow \theta =$ **54.41°**

Exam tip: First check the point lies on both surfaces: $4 + 1 + 4 = 9$ and $4 + 1 - 3 = 2$. ✓

Q10 **Answer: 440.3 (accept 437.7 to 442.9)**

Concept: The liquid that rises in the tube comes from the reservoir, whose level drops by L(a/A). The true vertical difference is $L \sin\theta + L(a/A)$.

$\Delta h = 0.15 \times (\sin 20° + 0.01) = 0.15 \times 0.35202 = 52.80$ mm

$p_1 - p_2 = \rho g \Delta h = 850 \times 9.81 \times 0.05280 =$ **440.3 Pa**

Exam tip: Ignoring the reservoir drop gives 427.8 Pa. Inclining the tube magnifies the reading by $1/\sin\theta \approx 2.9$ here.

Q11 **Answer: (D) 26.57°**

Concept: In rigid-body linear acceleration the free surface is perpendicular to the effective gravity, so $\tan\theta = a_x/(g + a_z)$.

$\tan\theta = 4.905/9.81 = 0.5 \rightarrow \theta =$ **26.57°**

The surface falls towards the front (direction of acceleration) and rises at the rear.

Exam tip: 63.43° is $\tan^{-1}(g/a)$, the angle of the effective gravity with the horizontal, not of the surface.

Q12 **Answer: 2.25 (accept 2.24 to 2.26)**

Concept: $h_{cp} = \bar{h} + I_G/(A\,\bar{h})$ for a vertical plane; for a triangle $I_G = bh^3/36$ about its centroid.

Centroid is $2h/3 = 2$ m below the apex (the free surface): $\bar{h} = 2$ m, $A = 3$ m²

$I_G = 2 \times 27/36 = 1.5$ m⁴ $\rightarrow h_{cp} = 2 + 1.5/(3 \times 2) =$ **2.25 m**

Total force $F = \rho g A \bar{h} = 9810 \times 3 \times 2 = 58.86$ kN

Exam tip: With the apex down and base at the surface, $\bar{h} = 1$ m and $h_{cp} = 1.5$ m. Always measure $\bar{h}$ from the free surface.

Q13 **Answer: (A) 2xy**

Concept: Integrate the two velocity components into ψ and match the arbitrary functions.

$u = \partial\phi/\partial x = 2x = \partial\psi/\partial y \rightarrow \psi = 2xy + f(x)$

$v = \partial\phi/\partial y = -2y = -\partial\psi/\partial x = -2y - f'(x) \rightarrow f'(x) = 0$, so $\boldsymbol{\psi = 2xy}$

Exam tip: φ and ψ of a potential flow are harmonic conjugates: the lines φ = const and ψ = const cut at right angles. Some books use $u = -\partial\phi/\partial x$, which flips the sign; read the convention given.

Q14 **Answer: 655 (accept 648 to 662)**

Concept: Continuity on the falling level with Torricelli's outflow: $-A\, dH/dt = C_d\, a\, \sqrt{(2gH)}$. Integrating gives $t = 2A(\sqrt{H_1} - \sqrt{H_2})/(C_d\, a\, \sqrt{(2g)})$.

$A = \pi(1.5)^2/4 = 1.7671\ m^2$, $a = \pi(0.05)^2/4 = 0.001963\ m^2$

$t = 2 \times 1.7671 \times (2 - 1)/(0.62 \times 0.001963 \times 4.429) =$ **655 s** (about 10.9 min)

Exam tip: Draining the last 1 m takes as long as falling from 4 m to 1 m, because the time depends on $\sqrt{H}$ differences.

Q15 **Answer: 17.89 (accept 17.82 to 17.96)**

Concept: For steady flow only the convective acceleration remains: $a_x = u\, \partial u/\partial x + v\, \partial u/\partial y$, $a_y = u\, \partial v/\partial x + v\, \partial v/\partial y$.

At (1, 2): $u = 4$, $v = -4$; $\partial u/\partial x = 2y = 4$, $\partial u/\partial y = 2x = 2$, $\partial v/\partial x = 0$, $\partial v/\partial y = -2y = -4$

$a_x = 4 \times 4 + (-4) \times 2 =$ **8 m/s²**; $a_y = 0 + (-4)(-4) =$ **16 m/s²**

$|a| = \sqrt{(64 + 256)} =$ **17.89 m/s²**

Exam tip: Check continuity first: $\partial u/\partial x + \partial v/\partial y = 2y - 2y = 0$, so the field is a possible incompressible flow.

What next?

Day 30 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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Next, Day 31/90: Fluid Mechanics, Momentum, Flow Measurement & Dimensional Analysis. Comment **GATE90** on our reel to get all links.
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